

Assignment 1 : Supervised Machine Learning

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I. DECISION TREE METHOD

A. Results

The decision tree algorithm was trained using the *rtg A*, *rtg B*, and *rtg C* training dataset. The generated decision tree was constructed using entropy-based Information Gain, with a stopping threshold of $IG < 0.00001$. The overall training classification accuracy achieved was:

$$Accuracy_{rtgA} = 100\% \quad (1)$$

$$Accuracy_{rtgB} = 100\% \quad (2)$$

$$Accuracy_{rtgC} = 100\% \quad (3)$$

The classification accuracy was calculated using:

$$Accuracy = \frac{\text{Number of correctly classified instances}}{\text{Total number of instances}} \quad (4)$$

B. Minimal Entropy

For each dataset, the feature resulting in the minimum entropy was identified by calculating the weighted entropy after splitting the entire training dataset on each available feature. The feature with the lowest resulting entropy represents the split that produces the purest separation of class labels and therefore provides the greatest reduction in uncertainty.

For *rtg A*, Feature 0 produced the minimum entropy value of 0.285506 with an Information Gain of 0.116673. Therefore, Feature 0 was selected as the root splitting feature of the generated decision tree. For *rtg B*, Feature 3 resulted in the lowest entropy value of 0.900495 with an Information Gain of 0.053199, making it the most effective feature for the initial split. For *rtg C*, Feature 3 achieved an entropy value of 0.000000 and an Information Gain of 0.904381, indicating that this feature completely separates the classes in the training dataset.

These observations were made by comparing the entropy values obtained from all features. The feature with the smallest entropy (or equivalently the highest Information Gain) was considered the most informative feature and was chosen by the decision tree algorithm for splitting the dataset.

C. Unused Feature

The features utilized by the decision tree were different for each dataset. For *rtg A*, Feature 3 was not used in constructing the decision tree. This occurred because the other features provided sufficient Information Gain to separate the classes, allowing the tree to reach pure leaf nodes before Feature 3 was considered. Since the stopping criteria were satisfied once the nodes became sufficiently pure, Feature 3 did not contribute additional useful information for classification.

For *rtg B*, all available features were utilized in the construction of the decision tree. This indicates that each feature contributed useful information during the splitting process. The tree required all features to further separate the data until the stopping criteria were reached.

For *rtg C*, only Feature 3 was used in constructing the decision tree, while the remaining features were excluded. This was because Feature 3 provided a very strong separation of the classes, achieving an entropy of 0 and a high Information Gain of 0.904381. After splitting on Feature 3, the resulting nodes were already pure, meaning additional features were unnecessary. Therefore, the remaining features were not selected because they did not provide any further improvement to the classification.

Overall, features were excluded from the tree-building process when they did not provide enough additional Information Gain or when previous splits had already produced pure leaf nodes. The decision tree only selects features that reduce class uncertainty and ignores redundant or unnecessary attributes.

D. Rtg C Discussion

1) *Generalisation Ability of the Generated Tree:* The generated decision tree is not expected to perform well on unseen data. The tree structure indicates that Feature 3 alone is able to completely separate the training instances, resulting in a very simple tree where the split is entirely dependent on a single feature. Although this produces perfect classification performance on the training dataset, it may not generalise well to unseen data because the model relies heavily on Feature 3.

If new instances contain feature values that were not represented in the training dataset, the tree may not have sufficient information from other features to make a reliable prediction. Therefore, the tree structure suggests a risk of poor generalisation due to its dependency on a single feature rather than learning patterns from multiple attributes.

2) *Unused Features in Tree Construction:* For the rtg C dataset, only Feature 3 was utilized in constructing the decision tree, while the remaining features were not selected. This occurred because Feature 3 provided a very strong separation between the two classes, achieving an entropy of 0 and a high Information Gain of 0.904381.

Since splitting on Feature 3 resulted in pure leaf nodes, there was no need for the decision tree to consider additional features. The other attributes may contain useful information; however, their contribution was unnecessary after Feature 3 had already completely separated the training data. Therefore, these features were excluded because they did not provide additional Information Gain during the tree-building process.

3) *Potential Preprocessing Improvements:* Although a test dataset is not available, several preprocessing approaches could be considered to improve the generalisation ability of the generated tree. Firstly, Feature 3 could be investigated further because the tree's complete dependency on this single attribute may indicate that the dataset contains a dominant or potentially artificial feature. Removing Feature 3 before training could encourage the model to learn from other attributes and produce a more generalised decision boundary.

Additionally, feature selection techniques could be applied to identify and retain informative attributes while reducing dependency on a single feature. Tree pruning or increasing the Information Gain stopping threshold could also be considered to prevent the tree from creating overly specific splits based only on the training data.

These preprocessing steps may reduce overfitting and improve the ability of the decision tree to classify unseen instances.

REFERENCES

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